

DEBRE TABOR UNIVERSITY
COLLEGE OF BUSINESS AND ECONOMICS
DEPARTMENT OF MANAGEMENT

MODULE
FOR THE COURSE MANAGEMENT INFORMATION SYSTEM

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CHAPTER ONE

INTRODUCTION TO MIS

Unit description

This chapter deals with the importance of information system for management and the types of information system. To deliver these contents, active learning methods such as brainstorming, interactive lecture, and group discussion will be used. And also to assess students' achievement, continuous assessments such as quiz, test, class activities will be used

Q. What do you mean by the word management information system?

1.1 Definition and Importance of MIS

The Management Information System (MIS) is a concept of the last decade or two. There are many closely related definitions in use. The terms MIS is synonymously used with terms the Information System (IS), the Information and Decision System and the Computer based Information System.

MIS is defined as an integrated system of man and machine for providing the information to support the operations, the management, and the decision-making functions in organizations.

The MIS is defined as a system which provides information support for decision making in the organization.

The MIS is defined as an integrated system of man and machine for providing the information to support the operations, the management and the decision making function in the organization.

The MIS is defined as a system based on the database of the organization evolved for the purpose of providing information to the people in the organization.

The MIS is defined as a Computer - based Information System.

Though there are a number of definitions, all of them converge on one single point, i.e., the MIS is a system to support the decision making function in the organization. The difference lies in defining the elements of the MIS. However, in today's world MIS is a computerized business processing system generating information for the people in the organization to meet the information needs decision making to achieve the corporate objective of the organization.

Information system is any organized combination of people, hardware, software, communication networks, and data resources that collect, transforms, and disseminates information in an organization. MIS deals with information that is systematically and routinely collected in accordance with a well-defined set of rules. In other words, data collection is a planned activity for which resources are allocated and rules are defined.

In order to get a better grip on the activity of information processing, it is necessary to have a formal system which should take care of the following points:

- Handling of a voluminous data.
- Confirmation of the validity of data and transaction.
- Complex processing of data and multidimensional analysis.
- Quick search and retrieval.
- Mass storage.
- Communication of the information system to the user on time.
- Fulfilling the changing needs of the information.

The management information system uses computers and communication technology to deal with these points of supreme importance

Q. What are the basic importance of management information system?

Information system resources: Information system consists of five main resources: People, Hardware, Software, Data and Networks.

Information system activities: The basic information processing activities that take place in information system include input, processing, output, storage, and control activities.

Roles of MIS

The MIS plays the role of information generation, communication, problem identification and helps in the process of decision making. The MIS, therefore, plays a vital role in the management, administration and operations of an organization.

Q. Explain the significance of management information system

1.2 Subsystems Of MIS

The two approaches to define subsystems;

1. Organizational function subsystems:

- a) **Marketing information system**- sales forecasting, sales planning, and sales analysis.
- b) **Manufacturing information system**- production planning and scheduling cost control analysis.
- c) **Personnel information system**- salary administration, planning HR recruitment, performance analysis.
- d) **Accounting information system**- financial analysis, cost analysis, capital planning, etc.
- e) **Logistics information system**- planning and control of purchasing, inventories, distribution.
- f) **Financial Information systems**:- computer based financial information systems support managers in decisions concerning the financing of a business, and the allocation and control of financial resources within a business.
- g) **Top management system**- strategic planning, resource allocation.

2. Activity sub system

- a) **Transaction processing system**- processing of orders, shipments, and receipts.
Transaction processing systems are information systems that process data resulting from the occurrence of business transactions. A transaction is a record of an event, which the business must respond
- b) **Operational control system**- scheduling of activities and performance reports.
- c) **Management control system**- formulation of budgets and resource allocation.
- d) **Strategic planning system**- formulation of objectives and strategic plans

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- h) **Strategic planning system**- formulation of objectives and strategic plans

1.3 The Evolution And Characteristics Of The Information Age

1) The Evolution of the Information Age

Agricultural Age: The period up to the 1800s, when the majority of workers were farmers whose lives revolved around agriculture.

Industrial Age: The period from the 1800s to 1957, when work processes were simplified through mechanization and automation.

Information Age: The period that began in 1957, in which the majority of workers are involved in the creation, distribution, and application of information.

- Knowledge Workers: Workers involved in the creation, distribution, and application of information.

Table 1.1 The Evolution of the Information Age			
	AGRICULTURAL AGE	INDUSTRIAL AGE	INFORMATION AGE
Time Period	Pre-1800s	1800s to 1957	1957 to present
Majority of Workers	Farmers	Factory workers	Knowledge workers
Partnership	People and land	People and machines	People and people
Principal Tool	Hand tools	Machines	Information technology

Q. Discuss the evolution of information technology

2) The Characteristics of the Information Age

- An information-based society has arisen.
- Businesses depend on information technology to get their work done.
- Information technology provides the means to rethink/recreate/reengineer conventional business processes.
- Success in business is largely determined by the effectiveness with which information technology is used.
- Information technology is embedded in many products and services.
- Reengineering efforts to attain greater productivity:
 - ✓ Industrial Age - Division of Labor: Separation of work process into component task, with different workers specializing in each of the tasks.
 - ✓ Information Age – Teamwork, Interconnection, and Shared Information.

1.4. What is Information Technology

- Information technology (IT) is a technology which uses computers to gather, process, store, protect, and transfer information. Today, it is common to use the term Information and communications technology (ICT) because it is unimaginable to work on a computer which is not connected to the network.
- Information technology is a systemic study of artifacts that can be used to give form to facts in order to provide meaning for decision making, and artifacts that can be used for organization, processing, communication and application of information.
- A term used to refer to a wide variety of items and abilities used in the creation, storage, and dispersal of data and information. Its three main components are computers, communications networks, and know-how.
- **Data:** Raw facts, figures, and details.
- **Information:** An organized, meaningful, and useful interpretation of data.
- **Knowledge:** An awareness and understanding of a set of information and how that information can be put to the best use.

Characteristics of Information Technology

Information Technology has the following Characteristics:

- Acquisition, Storage, manipulation, management, transmission or reception of data or information.
- Real time access to information.
- Easy availability of updated data
- Connecting Geographically dispersed regions
- Wide range of communication media.

Reference:

1. James A. O'Brien ,and GeorgeM.Marakas (20011) *Management Information Systems*, 10th ed, McGraw-Hill Irwin, PP 1-25
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3. Kenneth C. Loudon and Jane P Loudon(1999)Essentials of management information system,3rd ed, prentice hall; Newjersy
4. Rainer and Cegielski (2011) introduction to information systems: supporting and transforming business3rded, john Wiley & sons Inc.
5. Robert C. Nickerson(1998) business and information systems, Addison Wesley educational publication

Sample Question

1. Define management information system and list its importance?
2. Describe how information system can support a business?
3. Describe various types of Information Systems.

CHAPTER TWO

FOUNDATIONAL CONCEPTS IN MIS

Unit description

This chapter deals about the concepts of data, information, Knowledge & wisdom and the business systems. To deliver these contents, active learning methods such as brainstorming, interactive lecture, group discussion and independent learning will be used. And also to assess students' achievement, continuous assessment such as quiz, test, assignment, oral questions will be used.

Q. What are the business and management functions?

2.1 Data, information, knowledge and wisdom

Data has commonly been seen as simple facts that can be structured to become information. Information, in turn, becomes knowledge when it is interpreted, put into context, or when meaning is added to it.

There are several variations of this widely adopted theme. The common idea is that data is something less than information, and information is less than knowledge.

Moreover, it is assumed that we first need to have data before information can be created, and only when we have information, knowledge can emerge.

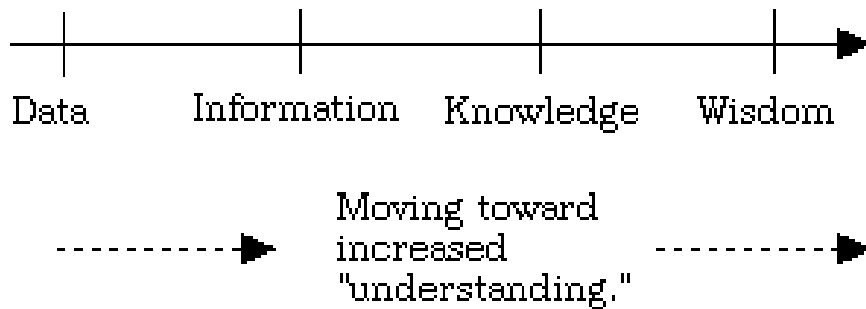
Data – raw facts and figures about people, places, events, and things.

Information – data that has been processed or reorganized into a more meaningful form for someone.

Knowledge – data and information that is further refined based on the facts, truths, beliefs, judgments, experiences, and expertise of the recipient.

Wisdom- the ability to make sensible decisions and judgments based on personal knowledge and experience.

Q. explain the difference between data, information and knowledge?



2.2 The information Needs and Sources of Managers

Who are the information users?

- **Managers:** From the managers' point of view, information serves the purpose of reducing uncertainty regarding the alternative course of action, in the process of decision making.
- **Non-managers:** Non-manages and staff specialists also use the MIS output.
- **Persons & organizations in the firm's environment:** Information generated by an organization can be valuable to many different people. They can be customers receiving invoices, stockholders getting dividend checks, and the federal government checking tax reports.

What are the sources of information?

An organizations information needs are met through gathering information from both internal and external environment. Internal information can be generated by maintaining and processing business transactional data and overall organizational data or information. To this end information system such as TPS & internal database systems play a critical role. While external information can be generated from diverse sources

2.3 A Framework For Information System

There is too much data and information in an organization. In order to design a MIS successfully, we need a framework to structure the information so that the data and information relevant for decision-making can be separated from rest of the data. For the successful designing of MIS in business, we should understand well about the type of information required by managers of the business.

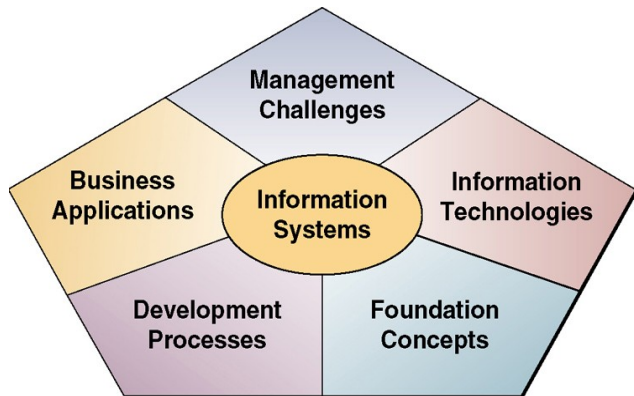


Figure 2.1: Information Systems Framework

Foundation concepts: The managerial end user needs to have knowledge of the generic components and properties of information systems, which requires an understanding of some basic concepts in general systems theory and information processing.

Development processes: How should managerial end users or information specialists develop information systems solutions to business problems? To answer this question, you should learn some basic problem solving and developmental concepts. One should understand how methodologies such as the systems approach, the systems development life cycle, and prototyping could be used to construct information systems applications that successfully meet end user and organizational needs.

Information Technology: In this area, they should have an understanding of major concepts, developments, and management issues in information technology, i.e. Hardware, Software, Telecommunications and Database management

Business Applications: They should gain a basic understanding of information systems concepts and applications in areas such as end user computing, office automation, transaction processing, information reporting, decision support, executive support and artificial intelligence.

Management challenges: This area requires understanding the major challenges and methods involved in managing the resources, technologies, and activities of information systems. More specifically, we should understand concepts such as information resource management and information systems planning, implementation, and control.

2.4 Business Systems

A business firm is an open, adaptive organizational system operating in a business environment. A business consists of the following interrelated system components:

- ❖ **Input:** Economic resources such as people, money, material, machines, land, facilities, energy, and information are acquired by a business from its environment and used in its system activities.
- ❖ **Processing:** Organizational Processes such as marketing, manufacturing, and finance transform input into output.
- ❖ **Output:** goods and services, payments to employees and suppliers, dividends, taxes, and information are all outputs produced by a business and exchanged with or transferred to its environment.
- ❖ **Feedback:** A primary role of information systems is serving as the feedback component of an organizational system. They provide information to management concerning the performance of the organization.
- ❖ **Control:** management is the control component of an organizational system.
 - ✓ Managers control the operations of a business so that its performance meets organizational goals such as profitability, market share, and social responsibility.
 - ✓ Feedback about organizational performance is compared to standards of performance established by management. Management then makes decisions to adjust performance to meet organizational goals.

2.4.1 E-business

Electronic business (e-business): All electronically mediated information exchanges, both within an organization and with external stakeholders supporting the range of business processes.

The Role of e-Business in Business

The Internet and related technologies and applications have changed the ways businesses operate and people work, as well as how information systems support business processes, decision making, and competitive advantage. Thus, many businesses today are using Internet technologies to Web-enable their business processes and create innovative e-business applications.

We define e-business as the use of Internet technologies to work and empower business processes, e-commerce, and enterprise collaboration within a company and with its customers, suppliers, and other business stakeholders. In essence, e-business can be more generally considered an *online exchange of value*. Any online exchange of information, money, resources, services, or any combination thereof falls under the e-business umbrella.

Q. discusses the reasons of using information technology in business activities?

References:

1. James A. O'Brien, George M. Marakas (2001) *Management Information Systems*, 10th ed, McGraw-Hill Irwin, PP 271-372
2. Robert Schultheis & Mary Sumner (1995) *Management Information Systems*, 3rd ed Irwin Pp 1-79
3. Kenneth C. Loudon and Jane P Loudon (1999) *Essentials of management information system*, 3rd ed, Prentice Hall; New Jersey
4. Rainer and Cegielski (2011) *Introduction to information systems: supporting and transforming business*, 3rd ed, John Wiley & Sons Inc.
5. Robert C. Nickerson (1998) *Business and information systems*, Addison Wesley educational publication

Sample Question

1. Describe the management functions at various levels in the context of relationships between management and information needs?
2. What are the functions that MIS supports in an organization?

CHAPTER THREE

INFORMATION TECHNOLOGY

Unit description

This chapter deals, the main information system resources, communication technologies and data base management systems. To deliver these contents, active learning methods such as brainstorming, interactive lecture, group discussion, independent study and the like will be used. To assess students' achievement, both continuous assessment such as quiz, test, assignment and final exam will be used.

Technology plays an important role in delivering timely and error free information to its

Q. What are the major components of information technology?

recipients. Technology includes hardware, software, databases, and communication system.

3.1 Hardware: Computing, Storing and communicating

Hardware is a set of devices such as processor, monitors, keyboard, and printer that accept data, process and display them.

A computer performs basically five major operations or functions. These are 1) it accepts data or instructions by way of input, 2) it stores data, 3) it can process data as required by the user, 4) it gives results in the form of output, and 5) it controls all operations inside a computer

Central Processing Unit: The ALU and the CU of a computer system are jointly known as the central processing unit. It is just like brain that takes all major decisions, makes all sorts of calculations and directs different parts of the computer functions by activating and controlling the operations.

Arithmetic Logical Unit (ALU): After you enter data through the input device it is stored in the primary storage unit. The actual processing of the data and instruction are performed by Arithmetic Logical Unit. The major operations performed by the ALU are addition, subtraction, multiplication, division, logic and comparison

Control Unit (CU): The next component of computer is the Control Unit, which acts like the supervisor seeing that things are done in proper fashion. The control unit determines the sequence in which computer programs and instructions are executed

Storage Unit: There are two kinds of computer memory: *primary* and *secondary*

- A. Primary Memory:** Computer memory is used to store two things: i) instructions to execute a program and ii) data when the computer is doing any job, the data that have to be processed are stored in the primary memory. This data may come from an input device like keyboard or from a secondary storage device like a floppy disk
- B. Secondary memory:** In this type of memory the cost per bit of storage is low. However, the operating speed is slower than that of the primary storage. Huge volume of data are stored here on permanent basis and transferred to the primary storage as and when required. Most widely used secondary storage devices are *magnetic tapes* and *magnetic disk*.

Input Output Devices: A computer is only useful when it is able to communicate with the external environment. When you work with the computer you feed your data and instructions through some devices to the computer. These devices are called Input devices. Similarly computer after processing gives output through other devices called output devices

Input Devices: Input devices are necessary to convert our information or data in to a form which can be understood by the computer. Keyboard, mouse and scanner are the most useful input devices

Output Devices: Visual display unit, printer and terminal are examples of output devices

3.2 programs: In Charge of the Hardware

Computer software: Computer cannot do anything without instructions from the user. In order to do any specific job you have to give a sequence of instructions to the computer. This set of instructions is called a computer *program*. Software refers to the set of computer programs,

procedures that describe the programs, how they are to be used. We can say that it is the collection of programs, which increase the capabilities of the hardware. Software guides the computer at every step where to start and stop during a particular job. The process of software development is called *programming*, and the individual who perform this task is called a programmer. There are two types of software: application software and system software

1. System software:

It consists of programs that control and direct the operation of computer hardware. Programs that manage and support a computer system and its information processing activities are system software. System software consists of programs that coordinate the various parts of computer system and mediate b/n application software and computer hardware.

Programs can be classified in to two broad categories: System management programs and system development programs.

A. Systems management programs: Systems management programs are programs that manage the hardware, software, network, and data resources of the computer system during its execution of the various information processing jobs of the users.

B. Systems development programs: Systems development programs are programs that help end users develop information system programs and procedures and prepare user programs for computer processing.

Operating systems: The most important system software package for any computer is operating system. The primary purpose of operating system is to maximize the productivity of computer system by operating it in a most efficient manner. Operating system performs five basic functions:

- Provide user interface
- Resource management
- Task management
- File management & Support services

2. Application software

Applications software are programs that specify the information processing activities required for the computing of specific tasks of computer user.

It also consists of programs designed for applying the computers to solve a specific problem.

It includes variety of programs that can be subdivided in to: *general purpose* and *application-specific programs*.

General- Purpose application programs are programs that perform common information processing jobs for end users

Applications – specific programs: Major categories of applications specific- programs include: Business applications programs- that accomplish the information processing tasks of important business functions

3.3 People: The Most Important Element

The effectiveness of an IT infrastructure is a function of the type of IT personnel available to it. Qualities like competence, honesty and reliability are critical to any work environment.

People resources include IS specialists and end users.

- IS Specialists – systems analysts, programmers, computer operators etc.
- End users – anyone else who uses information systems.

Q. distinguish between application and system software

3.4 Procedures: The Way It Goes

The procedures play an important role in a smooth and effective utilization of information resource. They also protect the information resource and maintain its quality. Thus the procedures as element of information system refer to the instructions to users regarding the use of IT infrastructure for normal day to day activity and also for handling special situations such as system failure and crashes. They include user access permissions and disaster recovery procedures as well.

Examples of procedures:

1. User instructions (for users of the application to record data, employ a terminal to enter or retrieve data, or use the result.)
2. Instruction for preparation of input by data preparation personnel.

3.5 Information: The Reasons for Using Information Technology

No matter how they may be classified, information systems have the following business values in an organization by supporting business operations, decision-making, and strategic management:

- Support of business operations.
- Support of managerial decision-making.
- Support of strategic competitive advantage.

3.6 Communication Technologies

Data Communication For communication of information and messages we use telephone and postal communication systems. Similarly data and information from one computer system can be transmitted to other systems across geographical areas. Thus data transmission is the movement of information using some standard methods. These methods include electrical signals carried along a conductor, optical signals along an optical fibers and electromagnetic areas

Communication Protocols The data communication software instructs computer systems and devices as to how exactly data is to be transferred from one place to another the procedure of data transformation in the form of software is commonly called *protocol*.

The data transmission software or protocols perform the following functions for the efficient and error free transmission of data.

1. Data sequencing: A long message to be transmitted is broken into smaller packets of fixed size for error free data transmission.
2. Data Routing: It is the process of finding the most efficient route between source and destination before sending the data.
3. Flow control: All machines are not equally efficient in terms of speed. Hence the flow control regulates the process of sending data between fast sender and slow receiver.
4. Error Control: Error

Data Transmission Modes: There are three ways for transmitting data from one point to another

1. Simplex: In simplex mode the communication can take place in one direction.
2. Half-duplex: In half-duplex mode the communication channel is used in both directions, but only in one direction at a time.

3. **Full-duplex:** In full duplex the communication channel is used in both directions at the same time.

Computer Network: *A computer network is an interconnection of various computer systems located at different places.* In computer network two or more computers are linked together with a medium and data communication devices for the purpose of communicating data and sharing resources. Some of the advantages of networks are:

- ❖ **Resource Sharing-** it is to make programs, files, equipment, and data available to anyone on the network without regard to physical location of the source and the user.
- ❖ **Provide High Reliability-** by having alternative sources of supply, it supports the critical function of backup.
- ❖ **Cost Minimizing-** by using small personal computers and file servers, we replace early expensive mainframes.
- ❖ **Flexible Working Environment-** employees can work at home by networking computers to their office,

Computer Networks may be classified on the basis of **geographical area** in three broad categories.

1. **Local Area Network (LAN):** Networks used to interconnect computers in a single room, rooms within a building or buildings on one site
2. **Metropolitan Area Networks (MANs):** are bigger versions of a LAN and normally uses similar technology. It might cover a group of nearby corporate offices or a city and might be either private or public
3. **Wide Area Network (WAN):** The term Wide Area Network (WAN) is used to describe a computer network spanning a regional, national or global area

Network Topology: The term topology in the context of communication network refers to the way the computers or workstations in the network are linked together. According to the physical arrangements of workstations and nature of work, there are three **major** types of network topology. They are **star topology**, **bus topology** and **ring topology**.

Q. Discuss the major contributions of computer networks

3.7 Database Management (DBM)

Data base: An integrated collection of logically related objects records using database management software.

DBMS: is a system software packages that helps you use the integrated collection of data records and files. It is a set of computer programs that control the creation, maintenance and use of databases of an organization and its end users.

The following are major conceptual categories of databases that may be found in many organizations.

i) *Operational database*: These databases store detailed data needed to support the business processes and operations of business enterprises. Store detailed data needed to support the operations of the entire organization. They are also called subject area database, transaction, and production data bases.

ii) *Analytical databases*: Store data and information extracted from operational and external databases, Summarized data & information most needed by managers & other end-users

✓ also called management databases / information databases

iii) *Data warehouse* data from current and previous years that has been extracted from the various operational databases of an organization

iv) *Distributed database* many organizations replicate and distribute copies of or parts of databases to network servers at a variety of sites. Such a distributed database is termed distributed database

v) *End-user database*: a variety of data files developed by end user at their workstations.

vi) *External database*: Access to wealth of information from external sources. (world wide webs).

Benefits and limitations of DB management

Advantage:

- Reduces duplication of data,
- Integrates data and
- Increase security of data

Disadvantage

- Difficulty and cost of installing

- Centralized data bases errors frond, and failures
- Decentralized database inconsistency

Q. what is data base management system?

Reference:

1. James A. O'Brien, George M. Marakas (2001) *Management Information Systems*, 10th ed, McGraw-Hill Irwin, PP 77-235
2. Robert schultheis & mary sumner (1995) *Management Information Systems*, 3rd ed Irwin Pp 90-225
3. Kenneth C. Loudon and Jane P Loudon (1999) *Essentials of management information system*, 3rd ed, prentice hall; Newjersy
4. Rainer and Cegielski (2011) *introduction to information systems: supporting and transforming business* 3rd ed, john Wiley & sons Inc.
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Sample question

1. What does general purpose software and special purpose software means?
2. What are the functions of operating system?

CHAPTER FOUR:

COMMON BUSINESS APPLICATIONS OF INFORMATION TECHNOLOGY

Unit description

This chapter deals with the common business applications of information technology. To deliver these contents, active learning methods such as brain storming interactive lecture, group discussion, and independent learning will be used. And also to assess students' achievement, continuous assessments such as quiz, test, class activities, assignments and others will be used.

Q. What are the common business applications?

4.1. Financial Information Systems

Financial accounting deals with collection, recording, and evaluation of financial data. Business enterprise requires systematic maintenance of their records that help for preparing the financial statements like Profit and Loss Accounts and Balance sheet. Accounting is considered as an information system because it has inputs of financial data, processing by evaluation and outputs through financial reports. The importance of financial information system for the different users in the managerial area can be depicted as follows.

Nature of Information	Ultimate User
Cost planning and cost control of operations Profitability of the firm Strategic and tactical decisions	Management
Profitability of the firm Shareholders and Soundness of the investment Growth prospects of the firm.	Investors
Liquidity of the firm Profitability and financial soundness	Creditors
Statement of salaries, wages and bonus	Employees
Managing the industrial economy of the country, collection	Government

of sales, excise and other taxes.	
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To fulfill the needs of financial information the information system used in most of the companies may be classified as under:

- 1) Financial Accounting Systems: To provide information to Government, investors, and other parties in the form of reports
- 2) Management Accounting System: To provide reports to managers for strategic and tactical decisions
- 3) Cost Accounting System: To provide information about cost planning and control operations to the managers

4.2 Marketing Information Systems

"A marketing information system is a continuing and interacting structure of people, equipment and procedures to gather, sort, analyze, evaluate, and distribute pertinent, timely and accurate information for use by marketing decision makers to improve their marketing planning, implementation, and control"

The explanation of this model of a marketing information system begins with a description of each of its four main constituent parts: the internal reporting systems, marketing research system, marketing intelligence system and marketing models.

4.4 Human Resource Information Systems

HR Information System: Let us discuss three of these subsystems separately:

Recruitment: Personnel department initiates process of recruitment, which is divided into the following steps:

- Finalizing the requirement for various posts
- Interviews
- Publicity as per requirement
- Final selection
- Receipt of application forms from candidates
- Issuing appointment letters
- Initial screening

Development proper information system giving the necessary inputs for generation of reports pertaining to the above cases lead to a subsystem for this purpose

Retention/Service Conditions: This is an important aspect in the functioning of personnel department. Personnel department maintains records of all the employees about the nature of the job, department etc. The major activities undertaken and related data requirements here are:

Training/ Seminars/Refresher, Performance Reports courses for employees, Salary/Compensation, Organizational Structure, Promotion, Job Description, Increments and others Information on each of the areas is compiled from the master database created and made available in the organization.

Retirement: Personnel department keeps track of retiring employees. It completes all the formalities necessary and issues orders for providing retirement package through account department. The calculation requires various inputs about the employees from the personnel records.

4.5 Managerial Decision Support Systems

A decision support system (DSS) is an interactive information system that provides information, models, and data manipulation tools to help make decisions in semi-structured and unstructured situations where no one knows exactly how the decision should be made

The Internet and Web have speeded-up developments in decision support and have provided a new means of capturing and documenting the development of knowledge in this research area.

Decision making is often seen as the center of what managers do; something that engages most of a manager's time. Decision making can be divided into 3 types: strategic, management control and operations control. Making decisions is not a single event but a series of activities taking place over time. Suppose, for example, that the Operations Manager for the National Milling Corporation is faced with a decision as to whether to establish buying points in rural locations for the grain crop. It soon becomes apparent that the decisions are likely to be made over a period of time, have several influences, use many sources of information and have to go through several stages. It is worth considering the question of how, if at all, information systems could assist in making such a decision.

4.6 Transaction Processing System (TPS)

Refers to the traditional applications of computers like invoicing, billing, order entry, dispatch, delivery, store accounting etc. In all these applications, any single transaction like raising an invoice, accepting an order, shipping an item, receipt of an item, payment of bill etc. would need an updating of multiple data stores (databases like accounts receivable, order status, pending list etc.)

Types of Transactions: When a department orders office supplies from the purchasing department, an internal transaction occurs, when a customer places an order for a product, an external transaction occurs.

Transaction Data Processing Methods:

- a. **Online transaction processing (OLTP):** is the almost instantaneous processing of data. The term online means that the input device is directly linked to the TPS and therefore the data are processed as soon as it is entered into the system.
- b. **Batch Processing:** in which transactions are accumulated over time and processed identically. Batch processing may be done on a daily, weekly, or monthly basis or any other time period appropriate to the application

4.7 Operation management information system

The area of management, which deals with the operational activities like production, project scheduling is called Operations Management. The important objectives of operations management are

- Inventory management
- Project scheduling
- Transportation costs minimization
- Assignment of jobs
- Sequencing of jobs
- Queuing

The problems in operations management are well structured. The tools of operations research need to be computerized for better results because of the following reasons.

- Easily structured through mathematical model
- Iterative in nature
- Numerical oriented

Reference:

Q. explains the difference between batch and online transaction processing?

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CHAPTER FIVE

KNOWLEDGE MANAGEMENT (KM)

Unit description

This chapter deals about knowledge management. To deliver these contents, active learning methods such as brainstorming, interactive lecture, group discussion, and independent learning will be used. And to assess students' achievement, continuous assessments such as quiz, test, class activities, assignments and others will be used.

How do you define knowledge management?

5.1 Introduction to Knowledge Management

Knowledge is like light. Weightless and intangible, it can easily travel the world, enlightening the lives of people everywhere. Yet billions of people still live in poverty unnecessarily. Knowledge about how to treat such a simple ailment as diarrhea has existed for centuries but millions of children continue to die from it because their parents do not know how to save them.

Knowledge informs decisions and actions. Knowledge is intangible, dynamic, and difficult to measure, but without it no organization can survive. Understanding Knowledge Management requires an understanding of knowledge and the knowing process and how that differs from information and data. **Knowledge Management** recognizes that data, information, and knowledge are critical organizational resources

Knowledge management is a discipline that promotes an integrated approach to identifying, managing and sharing all of an enterprise's information needs. These information assets may include databases, documents, policies and procedures as well as previously unarticulated expertise and experience resident in individual workers.” Source

Any **Knowledge Management** project should essentially have the following four components;

- ✚ Create knowledge (knowledge Repository)

- ✚ Improve Knowledge Access
- ✚ Enhance Knowledge Environment
- ✚ Manage Knowledge as an Asset

5.2 Sources and Types of Knowledge

“KM [Knowledge Management] involves blending a company’s internal and external information and turning it into actionable knowledge via a technology platform.”

The two sources of knowledge are:

Internal sources emerging from the operations of the organization-internal sources include the organizational operations such as design, development, engineering, sales, marketing, manufacturing, customer contact, etc. This is the basic source of organizational information, which is controllable and can be easily canalized to Knowledge Resource.

External sources such as Industry/Professional Associations, Commercial web sites etc-there are many professional bodies such as *Ethiopia Economic Association*, academic bodies such as universities, research institutions, industry associations, and commercial organizations.

There are two kinds of knowledge:

Explicit knowledge can be expressed in words and numbers and shared in the form of data, scientific formulae, specifications, manuals and the like. This kind of knowledge can be readily transmitted across individuals formally and systematically.

Tacit knowledge is highly personal and hard to formalize, making it difficult to communicate or share with others. Subjective insights, intuitions and hunches fall into this category of knowledge – Difficult to verbalize, such tacit knowledge is deeply rooted in an individual’s actions and experience, as well as in the ideals, values, or emotions he or she embraces.

Discuss the essential components of knowledge management?

5.3 KM sharing

We can share knowledge by converting the two types in different modes.

Knowledge creation in organizations takes place primarily through the dynamic process of four different modes of conversion between the two dimensions of knowledge.

Socialization: Tacit knowledge to conversion takes place when tacit knowledge within one individual is shared by another through training.

Combination: Explicit knowledge to explicit knowledge conversion takes place when an individual combines discrete pieces of explicit knowledge into a whole new.

Externalization: Tacit knowledge to explicit knowledge conversion can be said to take place when an individual is able to articulate the foundations of his and her tacit knowledge.

Internalization: Explicit knowledge to tacit knowledge conversion takes place when new explicit knowledge is shared throughout the firm and other members begin to use it to broaden, extend and reframe their own tacit knowledge.

Knowledge creation, whether for imitation or innovation, takes place at two levels- individual and organizational. The prime actors in the process of knowledge creation are individuals within the organization. Knowledge creation in organizations is not, however, the simple sum of knowledge creation by individuals.

5.4 Technologies to support KM

Knowledge as a product: Knowledge not only guides organizational actions, but can also be sold. For example, professional services companies sell knowledge services. Pharmaceutical companies hold *patents* and license the production of drugs.. Information and communication technology represents a key enabler for knowledge management initiatives.

Examples for information and communication technologies that are related to knowledge management are:

- 📁 *Intranet infrastructures* provide basic functionality for communication—email, teleconferencing—as well as storing, exchanging, search and retrieval of data and documents,
- 📁 *Document and content management systems* handle electronic documents or Web content respectively throughout their entire life cycle,
- 📁 *Workflow management systems* support well-structured organizational processes and handle the execution of workflows,
- 📁 *Artificial intelligence technologies* support for example search and retrieval, user profiling and matching of profiles, text and Web mining,

- ✚ *Business intelligence tools* support the analytic process which transforms fragmented organizational and competitive data into goal-oriented “knowledge” and require an integrated data basis that is usually provided by a data warehouse,
- ✚ *Visualization tools* help to organize relationships between knowledge, people and processes,
- ✚ *Groupware* and collaboration software supports for example the time management, discussions, meetings or creative workshops of work groups and teams,
- ✚ *E-learning systems* offer specified learning content to employees in an interactive way and thus support the teaching and/or learning process.

Discuss the four modes of converting knowledge?

5.5 KM application in organizations

Knowledge Management Initiatives at INFOSYS

Infosys provides consultancy and software services worldwide to 500 companies. KM at Infosys is founded on ‘Learn once and Use anywhere’ concept. Infosys manages organization-wide knowledge using three centrally operated knowledge Shops (k-shop); Process Assets Database (PAD) and People Knowledge Map (PKM).

Knowledge Management Initiatives at BAAN

Baan is a world leader in powerful, innovative, easy-to-use business software. Now they have turned towards knowledge management, in keeping with the demands of time. Two departments, namely, “Knowledge Transfer” and “Knowledge Development” are projected for this purpose. Their main objectives are to empower the member with skills necessary to meet the external world. One of the features of Baan is the encouragement provided to the employees for knowledge management.

Benefits of knowledge management

An effective Knowledge Management program should help a company leverage the assets and provide the following benefits:

- Fostering innovation by encouraging free flow of ideas.
- Improving customer service by streamlining response time.
- Boosting revenues by getting products and services to market faster.
- Enhancing employee retention rates by recognizing the value of employees knowledge and rewarding them for it.
- Streamlining operations and reducing costs by eliminating redundant or unnecessary processes.
- Enhance customer service and satisfaction by improving the speed and quality of customer contacts.
- Decrease customer service costs by improving self-service processes.
- Decrease training costs and times for new employees.
- Increase employee retention rates by recognizing and rewarding employee contributions.

Reference:

Q. What were the major contributions of Knowledge Management?

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Sample question

1. What are the benefits of knowledge management?
2. What are the problems & challenges in the implementation of a knowledge management system?

CHAPTER SIX

ETHICAL AND SECURITY ISSUES

Unit description

This chapter deals with the ethical and security issues to be considered while using information technology, to deliver these contents, active learning methods such as brainstorming, interactive lecture, group discussion, case study and independent learning will be used. And also to assess students' achievement, continuous assessments such as quiz, test, class activities, assignments and others will be used

Q. How do you think about the ethical issues in using IT?

The use of IT presents major security challenges, poses serious ethical questions, and affects society in significant ways. IT raises ethical issues in the areas of Crime, Privacy, Individuality, Employment, Health, and Working conditions. But, IT has had beneficial results as well

6.1 Ethics in information society

With their increasing power and versatility (**Functions**), computers simplify day-to-day life. Unfortunately, as computer use becomes more widespread, so do the opportunities for **misuse**. Computer hackers—people who illegally gain access to computer systems—often violate privacy and can tamper with(**interfere with**) or destroy records. Programs called viruses or worms can replicate (**duplicate**) and spread from computer to computer, erasing information or causing malfunctions(**fail to function normally**). Other individuals have used computers to electronically

embezzle(**steal**) funds and alter(**change**) credit histories. New ethical issues also have arisen, such as how to regulate material on the Internet and the World Wide Web.

Principles of Technology Ethics

Proportionality: The good achieved by the technology must outweigh (**greater than**) a harm or risk. Moreover, there must be no alternative that achieves the same or comparable benefits with less harm or risk

Informed consent: Those affected by the technology should understand and accept the risks.

Justice: Benefits and burdens of the technology should be distributed fairly. Those who benefit should bear their fair share of the risks, and those who do not benefit should not suffer a significant increase in risk

Minimized Risk: Even if judged acceptable by the other three guidelines, the technology must be implemented so as to avoid all unnecessary risk.

6.2 Types of threats to computer systems and control

Security concepts in a computer system

Security in a computerized system involves protecting all parts of the computer system. This includes data, the software and the hardware. Computer systems (hardware, software and data) need protector from intentional as well as unintentional damage and misuse.

Some scenarios of threats to computer system

- The system could succumb to fires (accidental. or deliberate) or earthquakes or floods dragging our equipment and making it impossible for the organization to function normally.
- Data in the system could be tampered with (**changed without Authority**)
- Confidential(**secured**) data may become available to unauthorized persons
- Software may get stolen and used by competitors.

Often, years of hard labor and creative effort goes into building of software, which may be a key to an organization's competitiveness

What is a Virus? Computer viruses are self-replicating (**coping themselves**) programs which cannot exist without a host. These programs can be relatively harmless, or can completely incapacitate (**prevent from functioning in a normal way**) your computer. The damage done to your computer is called the *payload*, and can range from a harmless message or image being displayed to destroying files and reformatting the hard drive of your computer.

With the popularity of the Internet, viruses have become more widespread and common. Originally, the viruses were passed by floppy disks, but now they are often passed through Internet downloads and especially email attachments. There are other types of programs associated with viruses and sometimes seen to be subsets of viruses. These are worms.

Protecting Your Computer from Viruses

With the prevalence(**wide spread**) of viruses today, it is important for any computer accessing the Internet to have protection against viruses. In addition to safe operating procedures, there are many different programs which can help you to protect your computer Trojans.

Some hints for preventing virus infections:

Don'ts

- Never use storage media from an untrustworthy source
- Never download files from a site or person that you do not trust
- Never open an email attachment from a person you do not know
- Never open an email attachment that looks unusual or suspicious

Do's

- Disable preview screens when using email
- Scan storage media (e.g., USB drives) before using them
- Use a real-time virus scanner or frequently check your system with an online virus scanner
- When using a real-time virus scanner, download virus updates every week
- Backup important files often
- Use personal firewall (**network designed to block unauthorized access**) software to prevent unauthorized Internet access to and from your computer

Personal firewall software controls what programs are allowed to access the internet, and prevents people from hacking into your computer.

6.3 Disaster recovery plans

Some of the more common means are:

- Locate computer installation away from electric transformers, boiler rooms, kitchens or any other equipment that can cause a fire
- Ensure that computer systems supporting mission critical application are housed in fire proof construction
- Take regular backups(**copy**) on tape/ floppies and store the backups in a fire safe. Also periodically, take backup copies off-site. Off-site means a location away from the location of the computer installation.
- Install special fire extinguishers for electrical and electronic equipment like inert gas fire extinguishers.
- Do not install water sprinklers near costly computer equipment as these can accidentally go off and damage the computers.
- Do not allow employees to smoke near the computers ensure that power supply equipment and wiring is done in a proper manner. Also ensure that appropriate fuse wires are installed at the right places in the power supply
- Install smoke detectors
- Ensure that the computer installation is relatively *free* of paper and other inflammable material
- Do not locate computer installations in the basement, as basements are typically the first places to get flooded

- As far as possible, avoid installing a computer system on the top floor of a building, as these are affected by seepage. In case *you* have no choice, ensure that the roof is treated against seepage.
- Do not install computers near underwater tanks, large drains, sinks, etc.
- Do not eat or drink near computer installations. The food or drink may spill over the equipment. Also, the smell of the food and drink may attract insects and rats, who can damage the equipment
- Install an air-conditioner to control the temperature and humidity.
- Restrict the entry of personnel to persons with badges and persons who are accompanied by an authorized employee
- If machines or terminals are provided to individuals and are not kept at a centralized place, encourage employees to use locks and keys to their rooms and machines.

The smart strategy is to be in pro-active mode rather than reactive mode.

Reference:

Q. explain the disaster recovery plans

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Sample question

1. Discuss the ethical issues involved in the information society.

2. Why is there a need to protect hardware and software from unauthorized access?